Three Objects, One Image: Why Depth Networks Distort the Impossible Triangle

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Modern Computer Vision, Technion • Code: https://github.com/Itamarshalom/three-objects-one-image

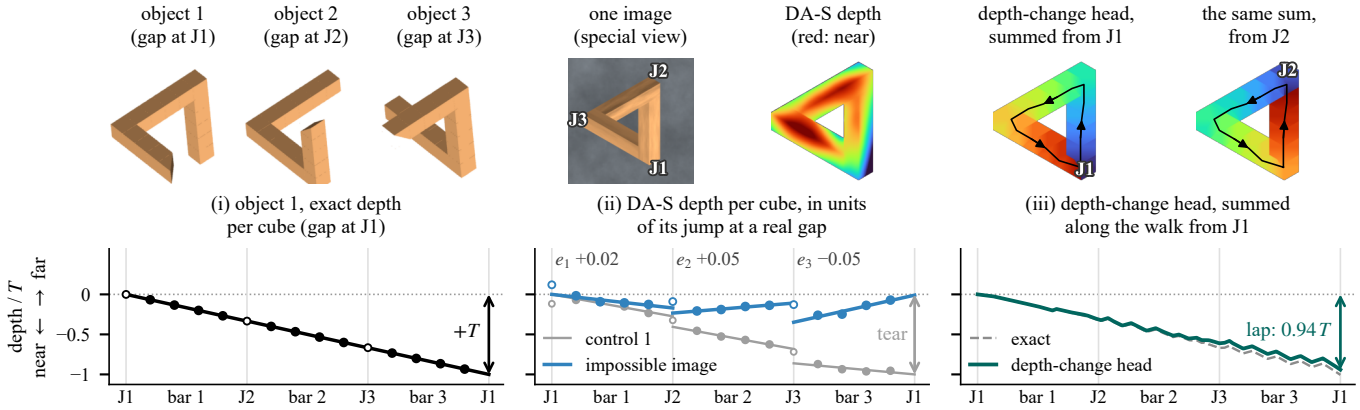


Figure 1: Three objects, one image. Top: three valid objects (turned 12° so their gaps show) give one image in the special view. Next to the image: the depth of Depth Anything V2-S (DA-S). Right: the output of a depth-change head on frozen DA-S, summed along the black walk from J1 or J2. Where the walk starts, the sum jumps by its total over one lap, as at an object's gap. Bottom: depth/T over one lap, up is far (hollow: junction cubes). (i) Object 1 comes nearer along each bar and goes back by T at its gap. (ii) DA-S makes only its ordinary step at each junction (excess tear $e_j \approx 0$), so its bars flatten or reverse and the loop closes. Grey: control 1 (object 1, gap visible), torn at J1. (iii) After one lap the head's sum is $0.94\,T$ nearer (dashed: exact). Its median over the impossible images is the same. Over the bars alone its median is $\Gamma = 0.96\,T$.

Abstract

Depth networks distort the Penrose triangle. We show that they do so because they must output a depth map, and that a linear read-out of Depth Anything V2 gets the figure's corners and bars right. We build the triangle from cubes, so its image is pixel for pixel the image of three valid objects, each with its gap hidden at a different corner. Ten depth networks output none of them and bend the bars to flat or reversed slopes. The read-out, fitted only on possible scenes, keeps each bar's slope until the last decoder stages. In this read-out every bar comes nearer, by a total T over one lap. It is known that a depth map cannot represent this, since depth must return to its start after a lap. Erasing one corner, or only blocking attention to it (on the cubes erasure leaves visible), removes most of the bend. Buildable rings of the same bars are mostly read correctly. We also train two heads on the same frozen features and scenes. The depth head bends the bars, as the networks do. The depth-change head keeps them. Its bars come $0.96\,T$ nearer per lap (0.00 on rings), and a cut at any corner gives the object with its gap there.

1 Introduction

Each corner and bar of an impossible figure is a valid piece of a solid, and the conflict appears only around the loop [4, 12]. In one special view, an open chain of three bars of cubes looks like the Penrose triangle, with its gap hidden at one of its corners (junctions J1–J3). If the bars are straight, the image has three valid readings, one per junction (Fig. 1). Other solids give it too [26]. Internally, the depth networks we probe read this figure as people do [16, 19]. They read every junction as joined and every bar as coming nearer around the loop. These depth changes (a 1-form) sum to zero locally and to a total T around the loop (in topology, closed but not exact [6, 20]). Depth must return to its start after a lap, so a depth map must cut the loop or bend the bars. A cut is one jump of T (a tear), as at each valid object's gap. A bend spreads the mismatch over the bars, and all ten networks we test do this.

It is known that no depth map can represent an impossible object [2, 6, 20]. Our three tests show that trained networks fail for this reason. A linear read-out of Depth Anything V2's features keeps every bar's slope until the last decoder stages. In all ten networks, erasing one junction removes most of the bend. On one frozen encoder with only possible training scenes, changing only the output from depth to depth changes keeps every bar's slope. Also, a network's depth is close to a least-squares fit of its local readings.

Related work. Meschers [6] stores an impossible object as a mesh 1-form with a nonzero sum around the loop. It builds the object cut or bent (by least squares) and fits one to a Penrose triangle image. Its authors argue that people see such objects locally. Akleman et al. [2] note that depth maps cannot represent such shapes and non-conservative fields can. Depth networks have been probed with image edits [27], frozen-feature read-outs [5, 7] and illusions [18, 29]. Jiang et al. [13] tested a depth network on impossible figures qualitatively. CNN classifiers fail to tell possible from impossible objects [11].

2 Method

2.1 Stimuli and measures

The loop is a chain of 15 unit cubes, five steps each along $+x$, $+y$ and $+z$. Seen orthographically along $v = (1,1,1)/\sqrt{3}$,

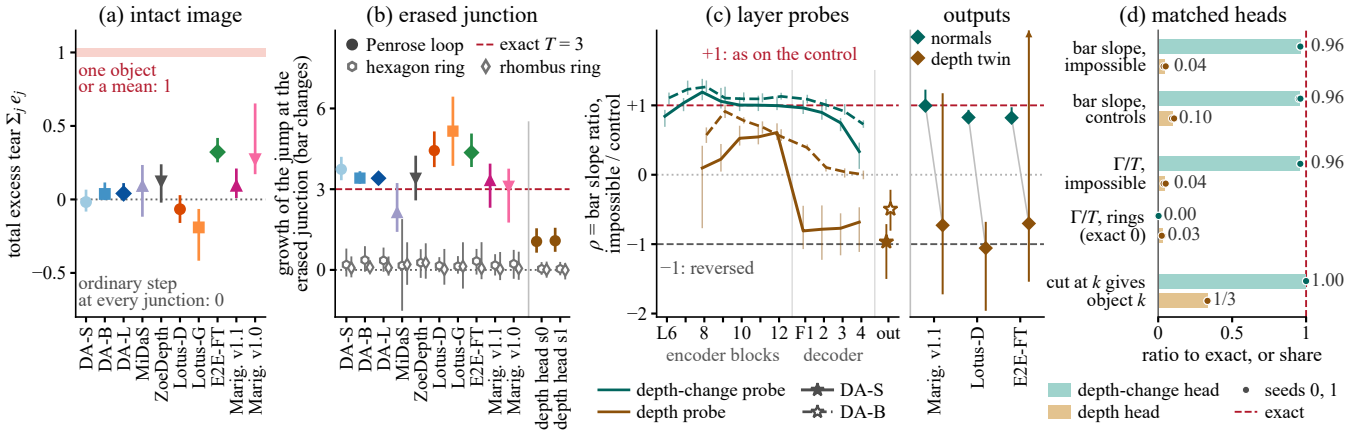


Figure 2: Results on the impossible loop. (a) Total excess tear $\sum_j e_j$ on the intact impossible image. One object, or any mean of objects, gives 1. An ordinary step at every junction gives 0. The networks give -0.19 to 0.32. **(b)** Growth of the jump at a junction when we erase it, in bar changes. It is large on the Penrose loop (filled) and near 0 on closed rings (grey, hollow). Dashed: $T = 3$, the size of a whole tear. Right of the grey line: our depth head (seeds 0 and 1), whose jump also grows. **(c)** Slope ratio ρ (impossible over control) of linear probes by layer in DA-S (solid) and DA-B (dashed). The depth-change probe (teal) stays near $+1$ until the last decoder stages. The depth probe (brown) is lower, and in the decoder it flattens the bars (DA-B) or reverses them (DA-S). Before their first point (L6, or L8 for depth), the probes find under 0.15 of the exact slope even on controls. Stars: the outputs of DA-S and DA-B. Right: three surface-normal networks (teal) stay near $+1$. Their depth twins (brown) reverse the bars. **(d)** Matched heads on frozen DA-S. The rows give the bar slope over the exact slope (impossible images, controls), Γ/T (impossible images, rings) and the share of cuts at junction k that give object k. The exact value is 1 (dashed), or 0 on rings. The depth-change head is near exact in every row. The depth head is near 0 in the first four rows and, by construction, at 1/3 in the last. Bars and numbers are means of two seeds (dots). Points: medians with 95% bootstrap intervals over rotations (IQRs in (b)). Carets: intervals past the axis.

its end projects onto its start. Object k has its gap at junction k (corner cube c_k). Its cubes before c_k move by $5(1, 1, 1)$, along v. So the three objects give one image, and $d_k = d_1 - T\,\mathbf{1}[c < c_k]$, with d_k object k's depth, c the cube index and $T = 5\sqrt{3}$. At the gap we trim the nearer bar's end by a prism parallel to v, so the three images are pixel-identical and each depth map is exact. We render 288 impossible images (24 in-plane rotations 5° apart, × 3 shadings × 2 textures × 2 chiralities), 48 possible controls (object k untrimmed, gap visible) and 64 buildable closed rings of the same bars with every junction visible (skew hexagons, rhombi). Probes and heads are trained only on 1500 random possible scenes rendered like the stimuli, with one or two open chains of straight runs. They never contain the Penrose family or a hidden gap that looks like a junction, so every contact in the image is one in 3D.

A cube's depth is the median output over its pixels at least 3 px from its edge. We fit a line to each bar's four interior cubes. A bar has the right sign if its slope has the objects' sign. The slope ratio ρ, a bar's slope on the impossible image over that on its matched control, needs no calibration ($+1$ if unchanged, below 0 if reversed). The jump Δ_j at junction j is the next bar's line minus the previous one, at j. Most outputs are known only up to shift and scale, so we measure each jump against the network's own controls. Let r_j be its mean jump at j on the two controls where j is joined, and g_j its jump on the control whose gap is at j. The excess tear $e_j = (\Delta_j - r_j)/(g_j - r_j)$ is 0 for its ordinary step at a joined junction and 1 for its jump at a real gap. We also give jumps in bar changes, where one bar change is an output's mean depth change along one bar (in this unit, $T = 3$). Rings have no gap controls (no e_j), so Fig. 2b uses this unit. The loop integral Γ is how much nearer the three bars come over one lap, junctions left out, so it tests only the bars. For exact local depth changes it is T on the impossible image and its controls, and 0 on rings.

Proposition 1 (Least-squares fit on a loop)**.** *Fit one depth per cube to depth-change readings on the links of a loop of cubes, with weights $w_i > 0$. If the readings sum to m around the loop, the fit's error ϵ_i on link i is the share $(1/w_i)/\sum_l (1/w_l)$ of m.*

(Minimise $\sum_i w_i \epsilon_i^2$ subject to $\sum_i \epsilon_i = m$. This is standard in depth from normals, where sparse (L_1) estimators cut the loop [1, 8, 21, 24].) Correct readings give $|m| = T$ on the impossible loop and $m = 0$ on a ring. With equal weights every ϵ_i is $m/15$, the bent shape of [6], so a link that reads δ_i keeps $\delta_i - m/15$. If all links read the same, the bars are flat. If junction links read larger changes than bar links, bar links read less than $m/15$ and the bars reverse. An erased junction (zero weight) gets the whole tear. Any answer built from the objects cuts ($\sum_j e_j = 1$) but needs an accidental view [9].

2.2 Networks, probes and interventions

We run ten frozen networks: Depth Anything V2 S/B/L (DA-S/B/L) [28], MiDaS DPT-L [22, 23], ZoeDepth [3], and, on the Stable Diffusion 2 backbone [25], Lotus-D/G [10], Marigold E2E-FT [17] and Marigold v1.0/v1.1 [14, 15] (one denoising step). The generative ones run on 16 impossible images with several seeds. Marigold v1.1, E2E-FT and Lotus-D also come as surface-normal networks, each with a depth twin on the same backbone. A normal is the local surface tilt. Like depth changes, normals need not sum to zero around a loop. Ranges are over networks, and the repository lists each. *Erasure* replaces junction j (corner cube and adjoining bar ends) by the exact background. *Gluing* tests Prop. 1 on network readings. It fits 15 cube depths by unweighted least squares to

the mean cube-to-cube changes (in bar changes) of its outputs on the three erased images (open loops).

Layer probes are linear read-outs (ridge regressions) of the frozen tokens of DA-S and DA-B, at ViT blocks L0–L12 and DPT fusion stages F1–F4. The depth-change probe predicts the exact local depth changes, and the depth probe predicts depth. Each is fitted on 1200 of the scenes and calibrated by dividing its slopes and Γ by the share of the exact slope it recovers on the other 300. The *attention cut* stops every token of Depth Anything V2, in every encoder block, from attending to junction j's $\sim$88 tokens on the intact image. The decoder still uses them. The *matched heads* are DPT decoders on frozen DA-S with an occlusion-edge channel, trained on 1400 of the scenes (100 to validate, two seeds each). They share data and schedule and differ only in the output and its loss. The depth head outputs depth (its loss also matches depth changes), and the depth-change head outputs local depth changes. Both see the whole image. On a head's output, Γ skips the pixels the head marks as edges. A least-squares fit with the loop cut at junction k gives the head's candidate for object k.

3 Results

Network outputs. No network cuts the loop (Fig. 2a). The total excess tear on the impossible image is -0.19 to 0.32. The mean bar slope is reversed in 0.96–1.00 of impossible images. On controls it has the right sign in 0.92–1.00 for DA-S/B/L and Lotus-D and in 0.31–0.79 for the other six. The networks tell a valid object from the impossible image junction by junction. A control's gap shows because the near bar ends in front of the far one. All $\sim$3,500 pixels that differ from the impossible image lie at this gap. The networks put their largest jump there in 0.88–1.00 of controls. In the impossible image every junction looks joined, as in a valid object in the special view.

Erasure and closed rings. Erasing one junction of the Penrose loop grows the jump there by 2.13–5.16 bar changes ($T = 3$, Fig. 2b), and most bars regain the right sign (0.58–1.00). Since the erased image is pixel-identical to an erased control, the network cannot know which it sees. So the closed loop reverses the bars. The closed rings are mostly read correctly (0.66–1.00 of the bars). Erasing a ring junction grows its jump by only 0.02–0.37, so the renders do not cause the reversal.

Layer probes. In DA-S and DA-B the depth-change probe finds almost no bar slope before block L6 (Fig. 2c), so the network computes the slopes that the probe finds later. From L7 to F1 the probe keeps the bars near their control's slope ($\rho = 0.96$–1.27). Its Γ is 1.08–$1.28\,T$, as on the controls, and at most $0.31\,T$ on the rings. The probe bends the bars at F4. The depth probe bends them at L8–L12 ($\rho = 0.10$–0.91) and in the decoder flattens them (DA-B, $\leq$0.39) or reverses them (DA-S, -0.68 to -0.81). On bar pixels identical in the impossible image and its control, the tokens differ by 0.16–0.47 of their spread (L7–L12). There the depth-change probe stays within 6% of its value on the control (L8–L12). The depth probe falls to 0.28–0.69 of it. The surface-normal networks also keep the bars' sign ($\rho = 0.82$–0.99), but their depth twins reverse it.

Least-squares gluing. Gluing a network's outputs on the three erased images predicts its intact output with median Pearson $r = 0.65$–0.91 (eight networks). The mean of the objects gives 0.16–0.66, and the best object per image gives 0.50–0.76. We also tried junction weights from 0.01 to 100. Weight 1, the same as for bar links, fits best or within 0.002 of the best. In all eight, junction links read larger changes than bar links, so the bars reverse.

Attention cut. On the intact image we measure only the cubes that erasure leaves visible. The tokens of j still attend to all tokens. In DA-S/B/L the attention cut tears at j ($e_j = 0.92$–1.13, or 0.59–0.85 counting j's cubes), and the bars regain the right sign (0.74–1.00). At a joined junction of a control it gives only $e = 0.21$–0.35 (the disturbance from the cut). So the mean e of an image's three cuts separates impossible images from controls (AUC 1.00, with known junctions).

Matched heads. The depth-change head's bars sum to $\Gamma = 0.96\,T$ around the loop and to 0.00 around the rings (Fig. 2d). Cutting its loop at junction k gives the profile closest to object k in every image. For a depth map every cut gives the same profile, so its share is 1/3. The head does not show which junctions are joined. Its edge channel leaves the loop closed on all 48 possible controls too, so $|\Gamma|$ cannot tell them from impossible images (AUC 0.56–0.60). The depth networks tell them apart, as they find the controls' gaps. A depth map's changes sum to 0 around any loop, so a depth head can only replace the bend with a cut. Trained in-domain like the probes, our depth head is weak on bars even on controls (0.09–0.11 of the exact slope), and it flattens them. On the impossible image they keep 0.39–0.47 of their slope on its controls (depth-change head: 1.00–1.01). At an impossible junction it makes only its ordinary step (median largest e: 0.06, -0.08 for the two seeds), at an erased junction it tears ($e_j = 1.27$, 1.44), and it finds the gap in 0.75–0.83 of controls. DA-S's own decoder, on the same encoder, reverses the bars ($\rho = -0.97$, filled star in Fig. 2c).

4 Discussion

A linear read-out of a network can get every junction and bar of this figure right. No depth map can represent the figure, so a depth network must distort it. People also read the loop as closed [16] and still call the figure impossible [19]. Our depth-change head does the first. Its edge channel does not show which junctions are joined. A network that outputs depth changes and a map of where the surface is joined could do both.

Limitations. We test only orthographic renders of one figure at one mismatch size, out of distribution for every network. We use linear probes on two networks and heads on one backbone (two seeds).

Use of AI tools. Claude assisted us with parts of the code and with drafting the text; we checked and approved all of it.

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